

Exam 1 — Business Intelligence: Answer Key

Sebastián Muñoz-Herrera & Héctor Hormazábal

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1 Case 1: Pares & Álvarez

1.1 Exercise 1 — Variable Measurement Types

Variable	Main Type	Subtype	Justification
Project Type	Qualitative	Nominal	Categories (Residential, Commercial, Industrial) have no natural order; they simply label different project classes.
Quality Rating	Qualitative	Ordinal	Categories (Poor, Fair, Good, Excellent) have a clear ranking, but the distances between levels are not measurable or equal.
Duration Days	Quantitative	Continuous (Ratio)	Measured in calendar days; has a true zero and can take any positive real value within a plausible range.
Number of Workers	Quantitative	Discrete (Ratio)	Integer count of individuals assigned; a true zero is meaningful and fractional values are not possible.
Cost Overrun	Quantitative	Continuous (Ratio)	Monetary amount (CLP) measuring excess over budget; true zero indicates no overrun. Negative values violate the field definition.
Region	Qualitative	Nominal	Geographic labels (Santiago, Valparaíso, Concepción) with no inherent ordering between regions.
Inspection Failures	Quantitative	Discrete (Ratio)	Integer count of failed inspections; a true zero is meaningful (no failures recorded).

Table 1: Variable measurement types for the Pares & Álvarez dataset.

1.2 Exercise 2 — Data Quality Assessment

Dimension	Problem?	Where?	Potential analytical impact
Completeness	Yes	One record has no operator annotation and the source document in ProTrack has not been verified.	If the record contains missing or unreliable values, any analysis including it may produce biased estimates or incorrect group statistics.
Uniqueness	No	No duplicate records are reported.	—
Consistency	Yes	(a) Two records in <i>Quality Rating</i> use “GOOD” (uppercase) instead of the standard label “Good”. (b) One record in <i>Region</i> uses “Stgo.” instead of the standard label “Santiago”.	Software tools will treat these as distinct categories, creating spurious extra levels that distort frequency counts, grouping operations, and any model that encodes these variables.
Validity	Yes	One record in <i>Cost Overrun</i> = −150,000 CLP. The operational definition states the field must be ≥ 0 .	A negative value violates the domain constraint; descriptive statistics (mean, minimum) and regression coefficients will be distorted if the record is retained without correction.
Accuracy	Yes	Three records with <i>Inspection Failures</i> = 0 are flagged: they may represent blank fields that defaulted to zero rather than genuine zero-failure outcomes.	If these are actually missing values coded as zero, correlations and averages involving Inspection Failures will be artificially deflated, misleading any subsequent analysis.

Table 2: Data quality assessment across five dimensions.

1.3 Exercise 3 — Distribution of Number of Workers

From the descriptive statistics:

- **Mean = 26.1, Median = 20.5:** the mean substantially exceeds the median, a classic indicator of **positive (right) skew**.
- **Q1 = 14, Q3 = 34,** giving an IQR of 20. The bulk of projects employed between 14 and 34 workers.
- **Min = 7, Max = 95:** the maximum is extremely high relative to Q3, suggesting the presence of at least one high-leverage observation.
- **Standard deviation = 18.9,** which represents approximately 92% of the median value, confirming high dispersion relative to the centre.

The distribution is **right-skewed**: most projects use a relatively modest workforce, but at least one project (max = 95 workers) pulls the mean upward and likely constitutes a statistical outlier.

1.4 Exercise 4 — Outlier Detection in Duration Days

Using the standard **IQR fencing method**:

$$\text{IQR} = Q_3 - Q_1 = 187.5 - 107.5 = 80 \text{ days}$$

$$\text{Lower fence} = Q_1 - 1.5 \times \text{IQR} = 107.5 - 120 = -12.5 \Rightarrow \text{no lower outliers possible}$$

$$\text{Upper fence} = Q_3 + 1.5 \times \text{IQR} = 187.5 + 120 = 307.5 \text{ days}$$

The reported **maximum is 480 days**, which exceeds the upper fence by:

$$480 - 307.5 = 172.5 \text{ days}$$

This constitutes a **clear statistical outlier** on the upper end. No values fall below the lower fence. There is therefore **one anomalous observation**: a project whose total duration is far beyond the typical range of the remaining 29 records.

1.5 Exercise 5 — Outlier Handling Strategy

Identified outliers: Duration Days = 480 (confirmed above); Number of Workers = 95 (suspected, given the large gap between $Q_3 = 34$ and $\text{max} = 95$).

Recommended procedure:

1. **Investigate first.** Determine whether each outlier is a data-entry error or a genuinely extreme but valid observation. A 480-day project may legitimately represent a complex industrial contract.
2. **If the value is erroneous:** correct or remove the record after consulting the source system (ProTrack).
3. **If the value is legitimate:** do not remove it arbitrarily. Preferred alternatives are:
 - **Winsorization:** cap values at the upper fence (e.g., 307.5 days) to reduce leverage while retaining the record.
 - **Log transformation:** apply $\log(\cdot)$ to Duration Days and Number of Workers before modelling, compressing the scale and reducing the influence of extreme values.

Assumption: Because the dataset is small ($n = 30$), removing records has a disproportionate impact on statistical power. Winsorizing or transforming is therefore preferred over deletion unless the value is confirmed to be a recording error.

1.6 Exercise 6 — Correlation Interpretation

(1) Duration Days and Cost Overrun: $r = 0.91$

This is a **very strong positive linear correlation**. Projects that take longer tend to incur substantially higher cost overruns. The relationship is analytically consistent: longer projects accumulate additional labour costs, material price escalation, and overhead. With $r = 0.91$, schedule duration is one of the strongest predictors of budget deviation in this dataset.

(2) Number of Workers and Inspection Failures: $r = -0.41$

This is a **moderate negative correlation**. Projects employing more workers tend to record fewer inspection failures. A plausible interpretation is that larger, better-resourced teams are more capable of meeting regulatory and quality requirements throughout execution. However, the moderate magnitude ($|r| = 0.41$) indicates that other variables — such as project type or duration — likely confound this association, and causal claims should not be drawn from correlation alone.

1.7 Exercise 7 — Variable Treatment for Linear Regression

The target variable is the **Project Score** (continuous, 0–100).

Variable	Treatment	Justification
Project Type	One-hot encoding (2 dummy variables; drop one category as reference)	Nominal variable with 3 unordered categories. OLS requires numeric inputs; $k - 1$ dummies are created (e.g., Commercial and Industrial, with Residential as reference) to avoid perfect multicollinearity.
Quality Rating	Ordinal encoding (0 = Poor, 1 = Fair, 2 = Good, 3 = Excellent), or one-hot encoding if equal spacing is not assumed	Ordinal variable. If equal intervals between levels are assumed, integer encoding is parsimonious. If that assumption is questionable, one-hot encoding is the safer alternative.
Duration Days	Log transformation to address skew and outlier; standardisation recommended for comparability	Continuous ratio variable with confirmed right skew and an outlier at 480 days. A log transform compresses the scale and reduces undue leverage. Standardisation aids coefficient interpretation.
Number of Workers	Log transformation; standardisation recommended	Discrete ratio variable with marked right skew (mean $\gg$ median) and a suspected outlier at 95. Same rationale as Duration Days.
Cost Overrun	Correct or remove the negative value first; then log transformation and standardisation	Heavily right-skewed ratio variable. The negative record (−150,000 CLP) must be resolved before any transformation. A log transform is appropriate given the large positive skew and wide range of values.
Region	One-hot encoding (2 dummy variables; drop one category as reference)	Nominal variable with 3 unordered categories. Identical logic to Project Type; one region (e.g., Santiago) serves as the baseline.
Inspection Failures	Use directly as a numeric count after verifying the three zero values; standardisation recommended	Discrete ratio variable with a limited and interpretable range (0–14). After confirming or imputing the suspicious zeros, the variable can enter the model as-is. Standardisation is recommended so that coefficient magnitudes are comparable across variables.

Table 3: Variable treatments for inclusion in the OLS regression model predicting Project Score.

2 Case 2: Petroquim S.A. — Multiple Choice Answers

Q	Answer	Reasoning
1	b	$R^2 = 0.654$ and Adj. $R^2 = 0.649$. The small gap between the two indicates that the predictors add genuine explanatory power rather than merely inflating fit through additional parameters. Explaining 65% of variance is meaningful in an industrial context.
2	a	The F -statistic tests the joint null hypothesis that all slope coefficients are zero. A significant F (here $p = 4.95 \times 10^{-92}$) implies that <i>at least one</i> coefficient differs from zero; it says nothing about each coefficient individually.
3	c	Comparing the unstandardised coefficients: $X1 = 1.31$, $X3 = 14.04$, $X4 = 42.10$, $X5 = -2.68$, $X6 = 4.58$. $X4$ (Active Process Units) has the largest absolute coefficient (42.10 MWh per additional unit), holding all other variables constant.
4	b	$p = 0.220 > 0.05$, so $X2$ is not statistically significant at the chosen significance level. However, removing a predictor should also consider its theoretical relevance and the risk of omitted-variable bias before a decision is made.
5	d	Standard OLS interpretation: a 1°C increase in ambient temperature is associated with an average <i>decrease</i> of 2.68 MWh in daily energy consumption, holding all other predictors constant. The coefficient is highly significant ($p < 0.001$).
6	b	Jarque-Bera $p < 0.001$ rejects normality. Skewness = 1.232 (positive, not symmetric) and Kurtosis = 1.690 (below 3, hence platykurtic, not leptokurtic). The residuals are non-normal : positively skewed and platykurtic. Option (b) is the closest available description.
7	a	Breusch-Pagan statistic = 13.783, $p = 0.032 < 0.05$. We reject the null of homoscedasticity: the residual variance is not constant (heteroscedasticity is present).
8	b	Durbin-Watson = $1.864 \approx 2$, which provides little evidence of autocorrelation in the residuals.
9	d	All VIFs are below 1.02, indicating no multicollinearity between the predictors. The high Condition Number (1.36×10^4) most likely reflects large scale differences among predictors rather than true multicollinearity.
10	b	$\hat{Y} = -77.61 + 1.3088(100) + 0.0381(500) + 14.0405(10) + 42.1038(5) - 2.6845(20) + 4.5774(8)$ $= -77.61 + 130.88 + 19.05 + 140.41 + 210.52 - 53.69 + 36.62 \approx \mathbf{406 \text{ MWh}}$
11	c	$t = 22.31$ far exceeds any standard critical value. The coefficient of $X4$ is statistically different from zero at any conventional significance level.
12	b	$p = 0.116 > 0.05$: the intercept is not statistically significant at $\alpha = 0.05$. It should generally be retained in the model regardless, as removing it forces the regression through the origin and distorts all other coefficients.

Q	Answer	Reasoning
13	a	R^2 increases ($0.654 \rightarrow 0.657$) but Adj. R^2 decreases ($0.649 \rightarrow 0.647$). The penalty for adding a parameter outweighs the marginal gain in fit: X7 adds complexity without genuine predictive value.
14	c	Standardising predictors converts coefficients into standard-deviation units, enabling direct comparison of relative importance across variables. R^2 , F -statistic, and t -statistics remain unchanged.
15	d	$e_{\text{std}} = \frac{600 - 406}{52} = \frac{194}{52} \approx 3.73$ A standardised residual of ≈ 3.73 exceeds the conventional threshold of ± 3 and flags this observation as a potential outlier.
16	a	A U-shaped (non-linear) relationship between ambient temperature and consumption is best captured by including both X5 and X5² in the model. This adds a quadratic term without discarding the linear component.
17	b	$\Delta\text{AIC} = 4733.0 - 4732.5 = 0.5$. A difference below 2 AIC units is conventionally regarded as negligible: both models are essentially equivalent .
18	b	$\Delta\hat{Y} = 1.3088 \times 15 + 14.0405 \times 2 = 19.632 + 28.081 \approx \mathbf{47.71 \text{ MWh}}$
19	d	Normality is violated (Jarque-Bera $p < 0.001$) and homoscedasticity is violated (Breusch-Pagan $p = 0.032$). Nevertheless, the model has a highly significant F -statistic and a meaningful R^2 . These violations are addressable (robust standard errors, transformations) and the model can still be useful after correction.
20	b	$\text{MAE} = \frac{100}{5} = 20 \text{ MWh}$
21	c	$\text{MSE} = \frac{2250}{5} = 450$ $\text{RMSE} = \sqrt{450} \approx 21.2 \text{ MWh} \Rightarrow \text{closest option: } \mathbf{21.9 \text{ MWh}}$
22	b	$\text{MAPE} = \frac{3.57 + 3.92 + 6.49 + 5.00 + 2.22}{5} \approx 4.24\% \Rightarrow \text{closest option: } \mathbf{4.5\%}$
23	c	M2 achieves the lowest AIC (4732.1) with an Adj. R^2 of 0.647, nearly identical to M1 (0.649), while removing only the non-significant predictor X2. This represents the best trade-off between goodness-of-fit and parsimony.

Q	Answer	Reasoning
24	a	From M3 to M6, R^2 and Adj. R^2 decline steadily while AIC and BIC rise monotonically. This confirms that each removed variable (X3, X5, X6) contributed non-redundant explanatory power . X4 emerges as the single strongest predictor (M6: $R^2 = 0.501$ with X4 alone).
25	c	Training accuracy = 0.812 vs. Test accuracy = 0.675: a 13.7 percentage-point drop. This gap is the classic signature of overfitting — the model learned patterns specific to the training set that do not generalise to unseen data.
26	a	A 70/30 class imbalance biases the classifier toward the majority class (Low), inflating overall accuracy while degrading recall and precision for the minority class (High), as observed in the test confusion matrix.
27	d	A naive classifier that always predicts “Low” would achieve $91/126 \approx 72.2\%$ test accuracy — higher than the model’s 67.5%. This demonstrates that accuracy alone is a misleading metric under class imbalance.
28	c	The model shows acceptable training performance but suffers from overfitting, class imbalance, and poor minority-class detection on the test set (recall for High = $11/35 \approx 31\%$). Before deployment, strategies such as cross-validation, resampling (SMOTE or undersampling), and threshold tuning should be applied.

Table 4: Multiple-choice answers with reasoning — Petroquim S.A.